# Name the force and its source

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Identify forces on a chosen object and name each interaction partner.

**SUCCESS CRITERIA - I can...**

1. Name the chosen object and each force source.
2. Distinguish mass from weight with units.
3. Include only forces acting on the chosen object.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

A cart travels *west* at **12 m/s** and *west* at **4 m/s 3 s** later. Is it speeding up? **Find acceleration size** and direction.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the prediction

**Reference:** $F_g = mg$.
$F_g$ is weight; m is mass. Use g = **10 N/kg** for CPA. **Name the source of every** force.

**Worked reasoning:** A book rests on a table. Earth pulls the book down; the table pushes the book up. Each force label names both the force and the object producing it.

**Your turn:** 1. **Draw the book's force diagram**.
2. **Name the force** on Earth paired with the book's weight.

Add and label force arrows.

Book resting on a level table; draw forces on the book only.

**Marzano check - circle:** ① ② ③

## 03 Which object gets the force?

A book rests on a level table. **Draw its two vertical forces** and **name who exerts each**.

Add and label force arrows.

The dot represents the object. Chosen object: book

**Marzano check - circle:** ① ② ③

## 04 Mass and weight

A **2 kg** bag hangs at rest from a vertical rope.
1. Use g = **10 N/kg**.
2. **Draw the forces** on the bag, then **find its weight** and the rope tension.
3. **Label each force** with its source.

Add and label force arrows.

Chosen object: hanging bag.

**Marzano check - circle:** ① ② ③

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### 05 After the throw

A ball is moving upward after it leaves your hand. Ignore air resistance.
1. **Draw its force diagram** and **name the source of every** force.
2. **Explain why upward motion** does not require an upward force.

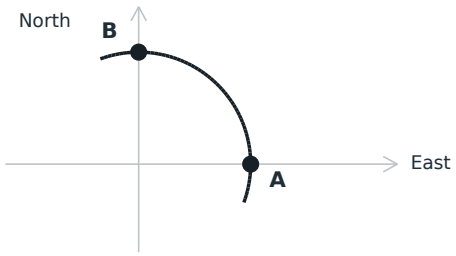
Add and label force arrows.

Ball after release; velocity is upward.

**Marzano check - circle:** 1 2 3

### 06 Review: connect an earlier idea

A rover travels straight *east* at a steady **5 m/s**. A second rover turns at the same steady speed. Which can have zero net force? Explain using velocity.
Use the curve for the turning case; **show how velocity changes**.




Path sketch: runner moves from A toward B.

**Marzano check - circle:** 1 2 3

### 07 Exit check

A **3 kg** lamp hangs still from one vertical cord.
1. Use g=**10 N/kg**.
2. **Name both forces** with source, magnitude and direction.

Add and label force arrows.

Hanging object at rest: label sources and magnitudes.

**Marzano check - circle:** 1 2 3

### 08 STRETCH: test the prediction

A hand presses down on the resting book. **Add the new force** and **explain how the table's support** must change.

Add and label force arrows.

Book pressed down by a hand: draw forces on the book.

**Marzano check - circle:** 1 2 3